Disaster Background

LITERATURE REVIEW

Post-Disaster Rescue Guidance via Fine-Tuning VLMs

Generate structured, rescue-actionable text from single post-disaster aerial & satellite images.

CSE499A · Section 15 · Group 5

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DisasterM3: A Remote Sensing Vision-Language Dataset for Disaster Damage Assessment and Response

WANG ET AL., NEURIPS 2025

- **Problem**: Generic VLMs struggle with disaster imagery due to domain gaps, sensor limitations, and lack of multi-task reasoning capability.
- **Objective**: Curate the first global-scale, multi-hazard, multi-sensor, multi-task remote sensing VLM benchmark for disaster damage assessment and response.
- **Scale**: 26,988 bi-temporal satellite images and 123,010 instruction pairs across 36 historical disaster events spanning 5 continents.
- **Three "M"s**: Multi-hazard (10 disaster types), Multi-sensor (Optical + SAR), Multi-task (9 distinct visual perception and reasoning tasks).

26,988
Bi-temporal satellite images

123,010
Instruction pairs

36
Historical disaster events
Across 5 continents

10
Disaster types
Multi-hazard coverage

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DisasterM3: 9-TASK TAXONOMY

From Simple Recognition to Complex Reasoning

DisasterM3 defines nine tasks across five functional categories, progressing from basic identification to actionable reporting.

Recognition

Disaster Scene Recognition, Disaster Type Recognition, Bearing Body Recognition — identifying *what* happened and *what* is affected.

Quantification

Damaged Building Counting, Damaged Road Area Estimation — measuring the *extent* of destruction using polygon masks.

Localization

Referring Segmentation — outputting precise bounding boxes/masks for specific damaged objects.

Reasoning

Object Relational Reasoning — describing spatial/causal relationships between damaged structures (e.g., "collapsed building adjacent to blocked road").

Reporting

Disaster Caption + Restoration Advice — generating comprehensive damage summaries and actionable Immediate/Long-term recovery plans aligned with UNITAR/FEMA guidelines.

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DisasterM3: METHODOLOGY & RESULTS

How They Built It, What They Found

Ground Truth Pipeline

GPT-4o generated question prompts → Human disaster experts annotated correct answers and drafted FEMA-aligned reports → GPT-4o polished drafts → Multi-round human verification.

Fine-Tuning

Trained Qwen2.5-VL-7B, InternVL3-8B, LISA, and PSALM on the 92K-pair Instruct Set.

Evaluation

LLM-as-a-Judge (GPT-4.1) scored open-ended outputs on a 1–5 rubric: Damage Assessment Precision, Factual Correctness, Action Priority Precision.

Headline Results

+10.4%

QA accuracy gain from fine-tuning

+2.1

Report score improvement

+40.8%

Referring Segmentation mIoU boost

Key conclusion: Domain-specific instruction tuning is **mandatory** — generic VLMs cannot achieve these results without targeted fine-tuning on disaster data.

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DisasterM3: Limitations, Gaps & Relevance to Our Project

What Remains Unsolved & Why It Matters for our project

snapshot of an pre-disaster image from xBD dataset

Unresolved Limitation: The Optical-vs-SAR cross-sensor performance gap remains significant even after fine-tuning — SAR imagery lacks the rich semantics of optical data.

Overfitting Risk: Fine-tuned counting models overfit on low-density scenes and degraded on high-density ones — a direct warning for our own pipeline.

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Research Gap: No existing benchmark combines multi-hazard, multi-sensor, and multi-task reasoning in one unified framework — DisasterM3 is the first attempt.

Relevance to Us: Their GPT-4.1-as-judge rubric scoring for reports directly justifies our planned RAR evaluation metric. Their use of Qwen2.5-VL + LoRA on xBD-derived data is architecturally identical to our own project's QLoRA pipeline on AIDER and xBD.

Representative: Abdullah Al Noman

From Pixels to Semantics: A Multi-Stage AI Framework for Structural Damage Detection in Satellite Imagery

Bijay Shakya, Catherine Hoier, Khandaker Mamun Ahmed

The Beacom College of Computer & Cyber Sciences

Dakota State University, USA

Representative: Abdullah Al Noman

The Three-Stage Pipeline

1. AI Super-Resolution

Enhances raw post-disaster satellite imagery from 1024 X 1024 px up to a razor-sharp 4096 X 4096 px using a specialized Video Restoration Transformer (VRT). This exposes tiny structural cracks and fine rubble details.

2. Building Detection

Deploys a fine-tuned YOLOv11 object detector on pre-disaster images to identify building footprints. It automatically crops each structure with a custom 30% safety padding to completely avoid cutting off edges.

3. Multimodal Analysis

Feeds pairs of cropped pre- and post-disaster crops to highly capable Vision-Language Models (VLMs). The AI classifies structural damage across 4 severity levels and instantly produces emergency safety recommendations.

Representative: Abdullah Al Noman

Key Findings & Validation

➤ Eliminating AI Bias with a Jury

- Instead of relying on a single AI's guess, the framework introduces a robust VLM-as-a-Jury panel to grade visual reports.
- Models evaluated include variants of Qwen3-VL (reasoning capabilities) and Gemma3 (non-thinking capabilities).
- Qwen3-VL 32B dominated the evaluation, achieving 87.1% accuracy and 0.87 F1-score on damage classification.

➤ Why Super-Resolution Matters

- Upscaling images with VRT directly boosted semantic alignment, yielding a solid +2.35 point increase in CLIPScore on Moore Tornado imagery.
- The framework is tested on actual real-world disasters from the xBD dataset, including the Moore Tornado and Hurricane Matthew.
- Provides direct, actionable insights to first responders, allowing them to instantly isolate highly compromised, unstable structures.

Representative: Abdullah Al Noman

LIMITATIONS & RESEARCH GAPS

➤ WHAT THEY HAVE YET TO ACHIEVE

- Fine-Grain Degradation: VLMs struggle to maintain high assessment accuracy when evaluating small, highly localized building crop patches.
- Visual Quality Collapse: Performance degraded consistently across all tested VLMs on cropped subsets due to low-quality, pixel-diluted visual features.
- Hallucination Risks: The framework has yet to fully resolve baseline model hallucination risks natively without relying on resource-heavy consensus.

➤ RESEARCH GAP IDENTIFIED

- Heavy Foundation Reliance: Current architecture relies on massive, computationally expensive foundation models operating in a zero-shot setup.
- Excessive Jury Overhead: Requires a complex, costly "jury" of multiple concurrent models just to verify and double-check output accuracy.
- Lack of Lightweight Alternatives: Clear research gap for a locally fine-tuned, smaller model (e.g., a 7B parameter backbone using QLoRA) to natively output structured data.

Representative: Tamanna Akter Mou

Satellite remote sensing and AI-driven analysis of post-disaster cultural landscape recovery

The Problem : Traditional cultural heritage management lacks rapid assessment capabilities in post-disaster scenarios where accessibility is restricted and time-sensitive interventions are critical.

Research Objective: To investigate the application of satellite remote sensing and AI in assessing, monitoring, and tracing the long-term recovery of 95 heritage sites globally (2018–2023).

Methodology: Integrated satellite-AI monitoring framework analyzing impact from earthquakes, floods, and wildfires across five global regions.

- Dataset of high-resolution UAV imagery captured after Hurricane Harvey.
- Focuses on urban flooding, building damage, and road traversability.
- Supports three core tasks: Classification, Semantic Segmentation, and Visual Question Answering.
- Critical for identifying flooded roads and natural vs. flooded water distinctions.

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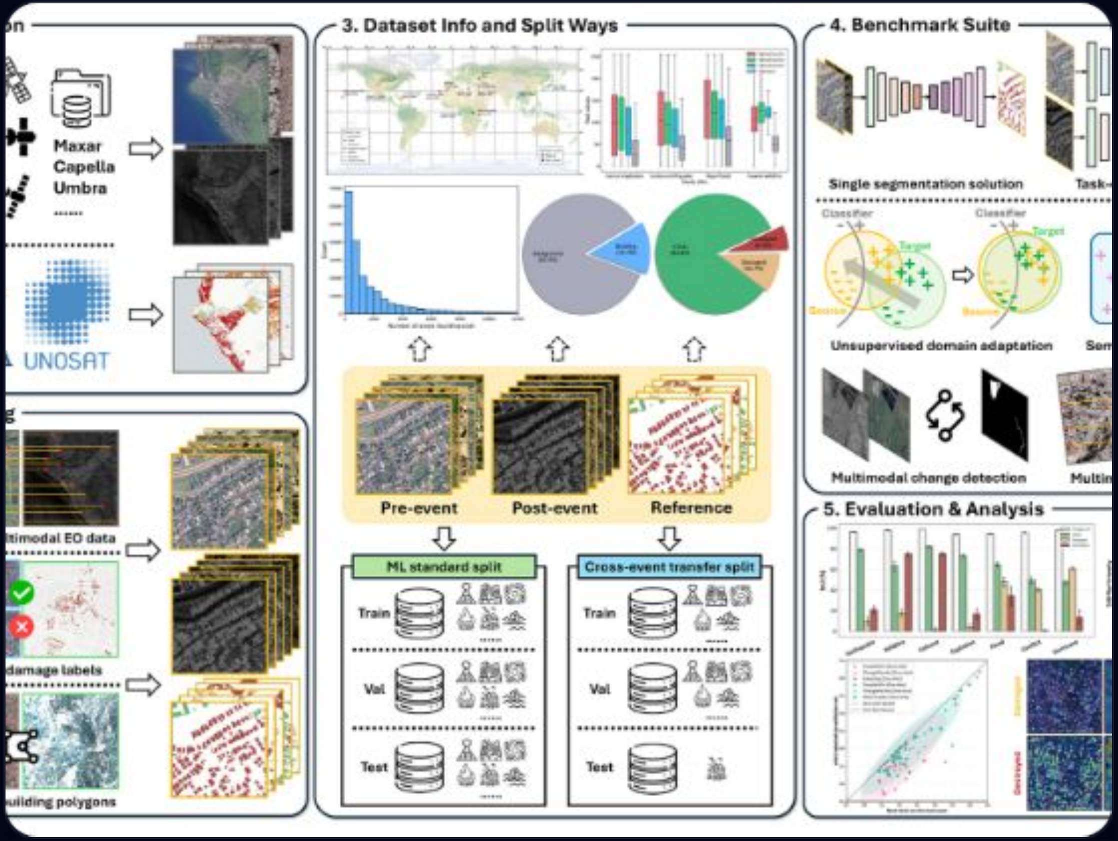
Representative: Tamanna Akter Mou

FLOOD SCENE UNDERSTANDING

- Provides clarity in scenes where satellite imagery fails due to cloud cover or smoke.
- Pixel-level annotations allow deep learning models to make precise damage assessment.
- Enables rapid evaluation of flooded buildings to prioritize water rescue teams.
- UAVs offer lower altitude shots, capturing survivor signals more effectively.



Landscape Scene Understanding

Key Findings & Performance Milestones

Accuracy Optimization

AI classification accuracy for landscape damage and recovery tracking surged from **75% to 95%** during the study period.

Global Initiatives

A significant increase in satellite remote sensing projects was observed, jumping from **4 to 33 active projects** globally.

Regional Focus

Analysis revealed the highest project concentration in **Asia (30)** and **Europe (25)**, aiding preservation decision-making.

Representative: Tamanna Akter Mou

Limitations & Project Relevance

What they yet to achieve:

Real-time interactive processing for immediate triage.Dynamic conversational querying for rescue coordinators.Direct generation of tactical, rescue-actionable text.

Research Gap : Existing literature remains **retrospective and passive**, focusing on quantification rather than providing situational directives during a crisis window.

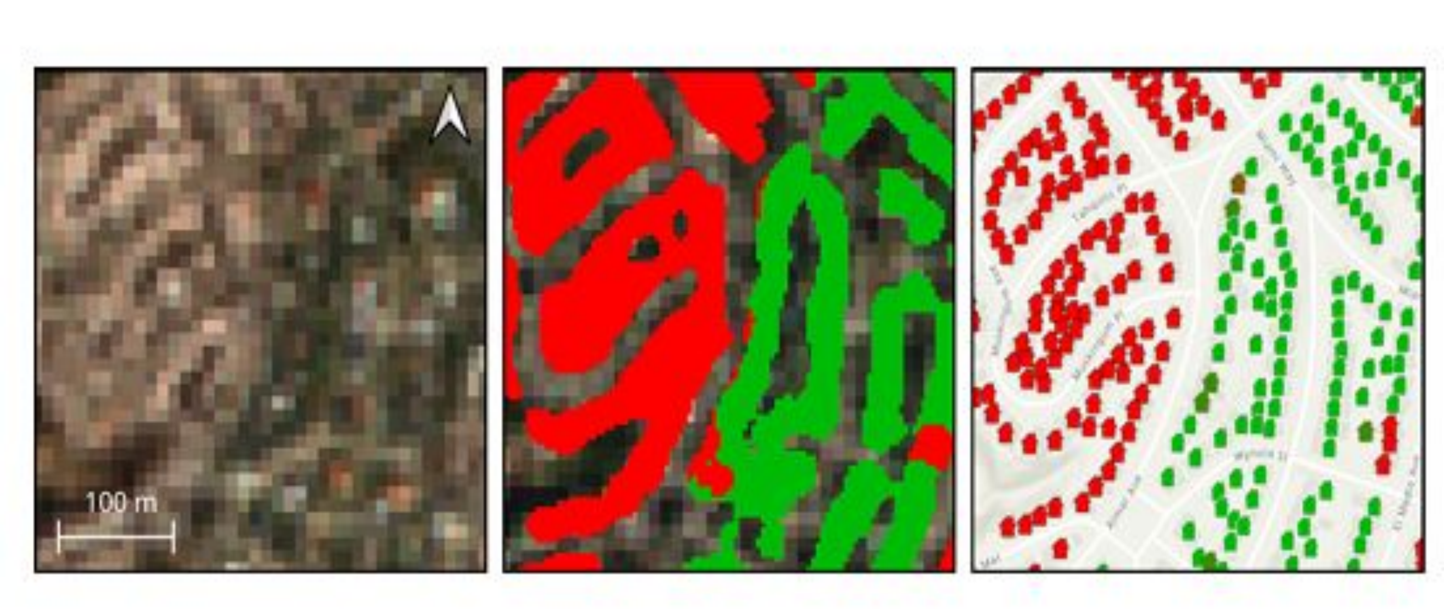
Relevance to Group 5: Our project bridges this gap by utilizing **Fine-Tuned VLMs (Qwen2-VL-7B)** to transform these high-accuracy visual patterns into active, text-based guidance for ground responders.

Representative: Aryan Sami

The Potential of Copernicus Satellites for Disaster Response: Retrieving Building Damage from Sentinel-1 and Sentinel-2

Dietrich et al., 2026

Using Sentinel-1 & Sentinel-2 for disaster response



Representative: Aryan Sami

Research Overview

Objective & Approach

- Test whether medium-resolution Copernicus satellite data (Sentinel-1 SAR, Sentinel-2 optical) can assess building damage during disasters
- Created xBD-S12 dataset by aligning Sentinel imagery with existing damage labels
- Trained multiple architectures including geospatial foundation models (Prithvi-EO-2.0, DOFA

Key Methods

- Corrected geolocation errors using building footprints
- Compared early vs. late fusion strategies
- Applied morphological buffers to handle coregistration errors
- Used ensemble methods and biased sampling for class imbalance

Results

- Evaluated using F1 scores for building localization and damage classification
- Tested on both random splits and geographically separated events
- Validated with real-world case studies (Mexico earthquake, Palisades fire)

Representative: Aryan Sami

Current Limitations

- Cannot detect sparse, low-intensity damage in dense urban areas
- Building localization remains weak at 10 m GSD
- No quantitative ground truth for real-world validation

Research Gaps

- Need for richer temporal information (longer Sentinel-1 time series)
- Incorporation of InSAR coherence or damage-proxy maps
- Further development of geospatial foundation models for fine-grained damage mapping

Relevance to Your Project

- Validates use of pre- and post-disaster satellite comparisons for identifying damaged areas
- Justifies incorporating multi-source remote sensing (optical + radar) for better disaster understanding
- Supports generating rescue-oriented image-response pairs from disaster imagery

Integrating Large Vision-Language Models for Post-Disaster Damage Assessment

Chen et al., 2026 — DisasTeller: A Multi-Agent LVLM Framework | Nature Communications

Related Work — how autonomous LVLM agents are reshaping emergency response

Group 5

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Representative: Ridita Afrin Riya

The Paper & Its Contribution

Objective:

introduce a multi-LVLM-powered framework that automates post-disaster management — on-site assessment, emergency alerts, resource allocation, and recovery planning.

Core Idea:

Four specialised LVLM agents (Expert, Alerts, Emergency, Assignment) — coordinated with GPT-4o as core — simulate real disaster-response teamwork.

4 Agents:

- Expert Team – On-site damage grading
- Alerts Team – Danger-zone alerts
- Emergency Team – Shelters & recovery
- Assignment Team – Resource allocation

Research Gap:

- No GIS Integration — spatial logic not linked to systems like QGIS
- Ungrounded Web Retrieval — depends fully on open web search, no curated knowledge base
- No Structured Validation — no formal ground-truth benchmark, only human-in-the-loop
- Limited to Zero-Shot Reasoning — no fine-tuning explored for disaster-specific tasks

Representative: Ridita Afrin Riya

Relevance to Our Project

DisasTeller shows LVLMs can ingest multi-source disaster imagery (local + map view) and output actionable reports without full fine-tuning.

DisasTeller:
- Zero-shot, prompt-based
- Crisis image dataset (ASONAM 17)
- RAG grounded in open web search
- No GIS-linked spatial reasoning

Our Project:
- Multi-agent LVLM ingestion of local + map-view images
- Zero-shot & Chain-of-Thought prompting with RAG
- Extends toward rescue-oriented reporting
- Targets the spatial-reasoning gap DisasTeller leaves open